

Subject Description Form

Subject Code	EEE522																											
Subject Title	Autonomous Vehicles																											
Credit Value	3																											
Level	5																											
Pre-requisite/ Corequisite/ Exclusion	Nil																											
Objectives	1. To acquire a high level of electric vehicles technology and future EV design. 2. To understand the development of the impact of autonomous vehicles on society and security.																											
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the advanced knowledge of the electric autonomous vehicle. b. Understand various sensors and components in autonomous vehicles. c. Assess the future development for autonomous vehicles. d. Explore impact of autonomous vehicles and emerging technologies.																											
Subject Synopsis/ Indicative Syllabus	1. Unmanned vehicle: History of unmanned vehicle, driver assistance functions, and the recent development. 2. Vehicle architecture: System design, hardware requirement, computer processing, end-to-end autonomous driving decision control. 3. Sensors: Sensors requirement, Lidar, Ultrasonic, visual camera, IMU, Radar, GPS. 4. Standard and infrastructure: Classification, SAE standard, connected vehicles, V2X, communication, application, Autonomous Ride Service (ARS), Economic and business model, Autonomous vehicles as a workspace. 5. Software for autonomous vehicle: Navigation and control, 3D-SLAM, HD-Mapping, routing, Motion planning, Perception System, Decision-making system, Digital recognition, Neural Network, Deep learning model. 6. Control: Electronic control systems, Kinematic model, Geometric control, Dijkstra's Algorithm, Stanley Controller, Convolution. 7. Legal and policy: Government Policy in autonomous vehicles, social cost of mobility, infrastructure of autonomous vehicles and the environment.																											
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures, complemented by tutorials, worked examples and assignment. Self-learning on the part of students is strongly encouraged and extensive use of web resources will be made. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Intended subject learning outcomes</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>1. Lectures</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>2. Tutorials</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> <tr> <td>3. Assignment/mini-project</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td></tr> </tbody> </table>				Teaching/Learning Methodology	Intended subject learning outcomes				a	b	c	d	1. Lectures	✓	✓	✓	✓	2. Tutorials	✓	✓	✓	✓	3. Assignment/mini-project	✓	✓	✓	✓
Teaching/Learning Methodology	Intended subject learning outcomes																											
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1. Lectures	✓	✓	✓	✓																								
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3. Assignment/mini-project	✓	✓	✓	✓																								

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Assignment/mini-project	25%	✓	✓	✓	✓
	2. Test	25%	✓	✓	✓	✓
	3. Examination	50%	✓	✓	✓	✓
	Total	100 %				
The assignment is designed to assess students' understanding of the advanced electric vehicle principles and its impact to society and whether they can present the study clearly. Oral presentation for their assignment is needed. It includes the take-home assignment and mini-project. The test is designed to assess students' understanding of the topics that they have learnt relative to learning outcomes (a), (b), (c) and (d). The test is usually conducted in the mid-semester to measure students' performance. Examination: questions are designed to assess learning (a), (b), (c) and (d). Students are required to answer questions that cover all of the learning outcomes.						
Student Study Effort Expected	Class contact:					
	▪ Lecture					30 Hrs.
	▪ Tutorial and presentation					9 Hrs.
	Other student study effort:					
	▪ Mini project or Assignment					27 Hrs.
	▪ Self-study					49 Hrs.
	Total student study effort					115 Hrs.
Reading List and References	1. Sreevatsan Bhaskaran, Kai Zhou , Andrew Baab, Ronald Calhoun, “Autonomous Vehicle Lidar: A Tutorial”, Independently published,2019. 2. Sumit Ranjan, S. Senthilarasu , “Applied Deep Learning and Computer Vision for Self-Driving Cars: Build autonomous vehicles using deep neural networks and behavior-cloning techniques”, Packt Publishing , 2020. 3. Paul McManamon, “LiDAR Technologies and Systems”, SPIE--The International Society for Optical Engineering, 2019. 4. Hanky Sjafr. “Introduction to Self-Driving Vehicle Technology”, Chapman & Hall/CRC Artificial Intelligence and Robotics Series, 2019. 5. S. Liu, L. Li, J. Tang, S.Wu, J.Gaudiot, “Creating Autonomous Vehicle Systems”, Synthesis Lectures on Computer Science, 2020.					